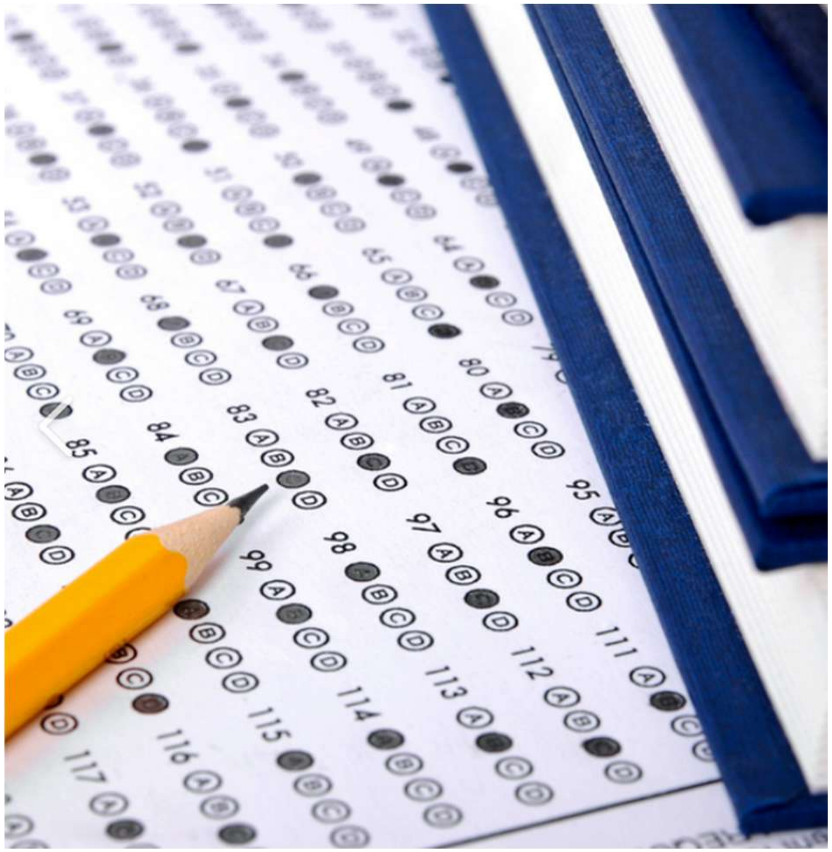
Exploring Trends in California Grade 4 Math Proficiency



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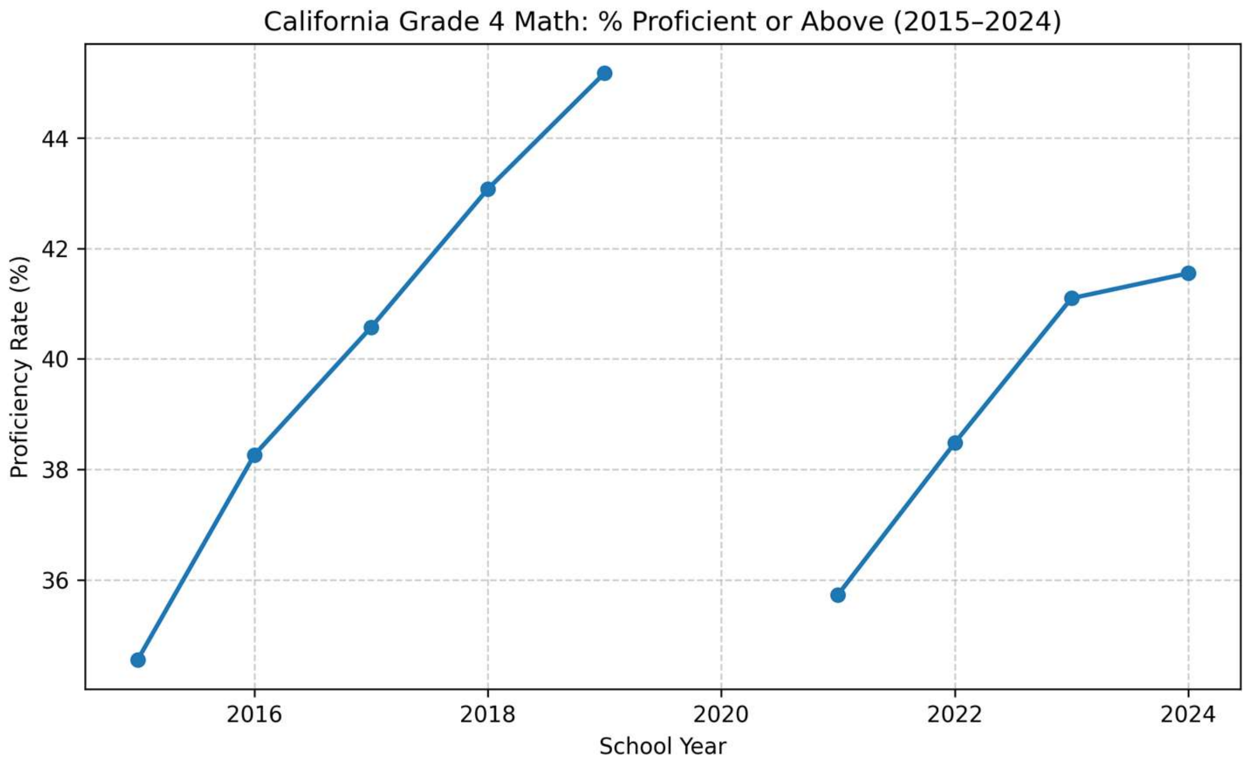
Repository Link: https://github.com/nseelig/DATA-1030-Final-Project.git

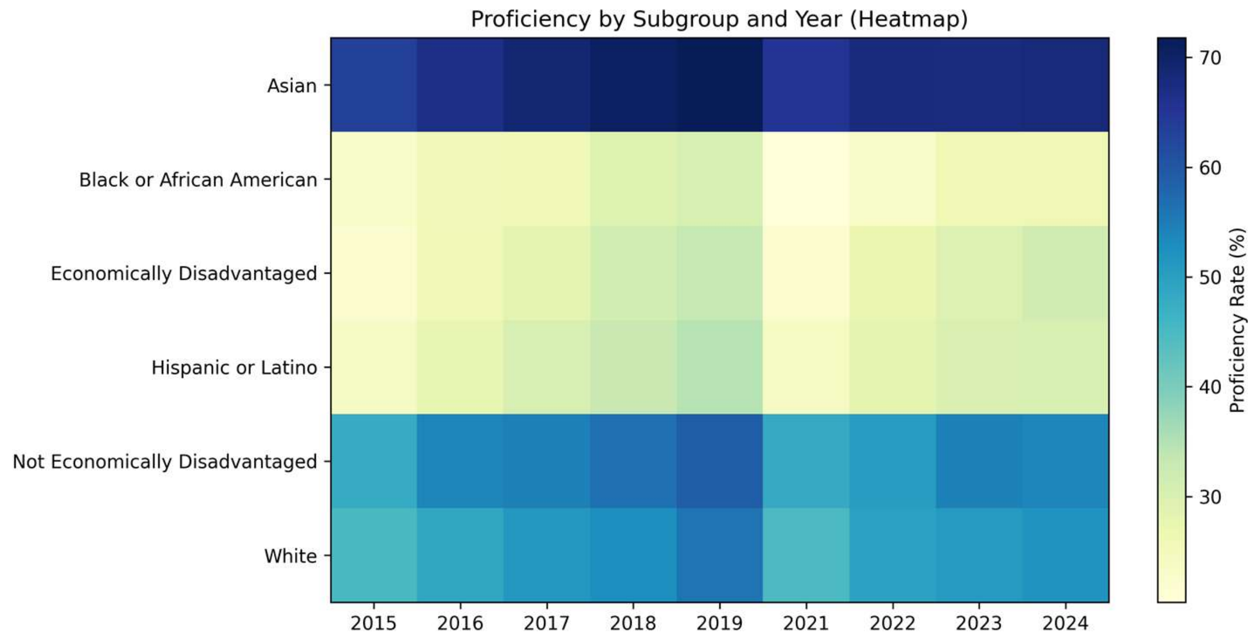
Introduction

- Examines math proficiency among California 4th-grade students between 2015 and 2024
- Data source: CA statewide assessment results (2015–2024, excluding 2020 pandemic year)
- Challenges
  - Missing data (about 30% of proficiency percentage values)
  - Non-IID structure

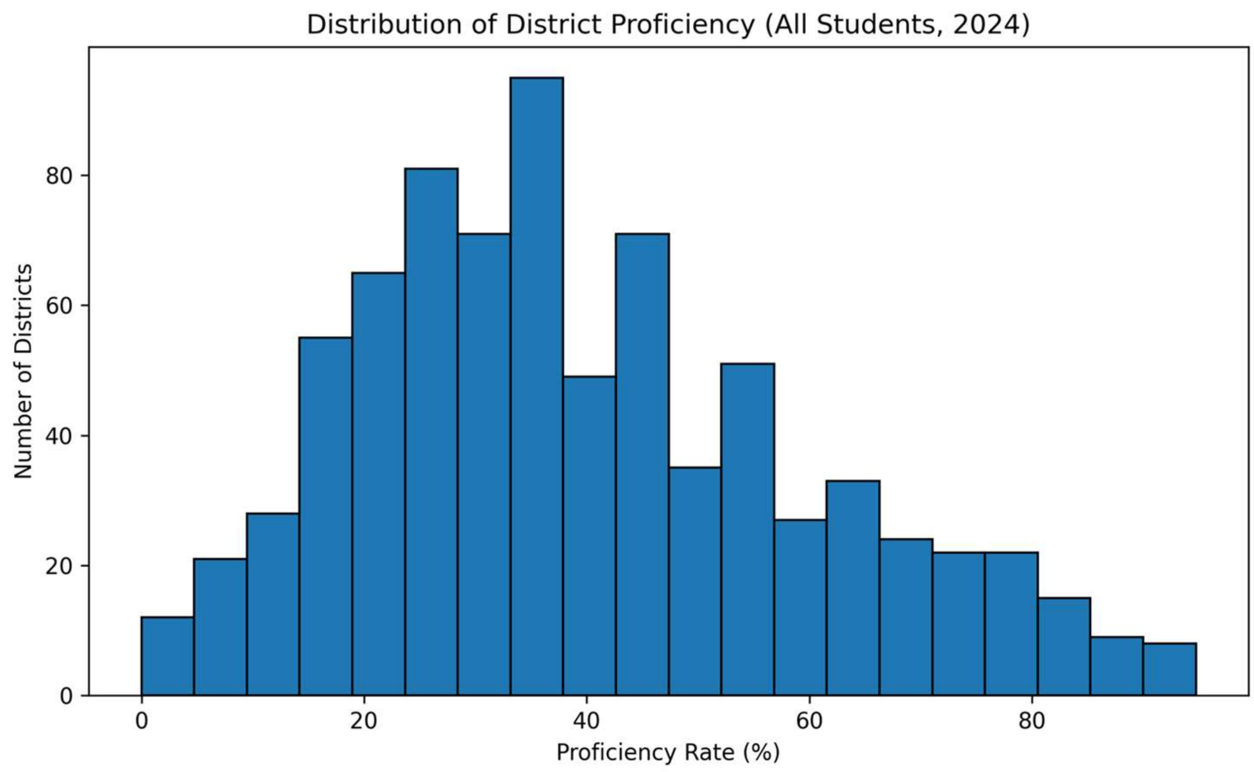
Aggregate Trends



Subgroup Differences



District Variation



Data Splitting

Unit: district x year

Target: percentage proficient and above

Features: participation rate, total tested, school year, student groups

Split: 80/20 train-test, stratified by year

Preprocessing

- Keep rows with valid target + non-zero tested
- Median impute missing numeric values
- Standardize numeric features, One-hot encode categorical features
- Result: ~7k rows × ~10 model features (after preprocessing)

Questions?